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Terrorist and Cartel Use of Unmanned Aerial Systems: A Comparative Assessment

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ABSTRACT

This paper compares unmanned aerial systems use by terrorist and transnational criminal organizations using ACLED and GRID data, supplemented by cross-regional case studies. We find that state-sponsored terrorist groups, particularly Iran-backed organizations, have advanced furthest along the technology adoption curve, integrating military-grade drones into precision strikes and combined-arms operations. In contrast, non–state-sponsored terrorist groups resemble Mexican cartels in their reliance on commercially available drones, decentralized supply chains and improvised modifications. Across actor types, drones are primarily used for intelligence, surveillance and reconnaissance and short-range explosive delivery. Terrorist drone attacks are deadlier overall, while cartels employ drones more frequently against civilian targets.

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Introduction

On 20 February 2025, the U.S. State Department officially designated eight international cartels and transnational criminal organizations (TCOs) as Foreign Terrorist Organizations (FTOs) and Specially Designated Global Terrorists (SDGTs). On the same day, the authors convened a practitioner focus group on the evolving threat of terrorist use of unmanned aerial systems (UAS, a.k.a. "drones"). Participants, including representatives from multiple U.S. federal law enforcement and homeland security agencies, noted that the designation fundamentally altered how policymakers and security practitioners should approach cartel drone activity. As one participant observed: "We've long been concerned about terrorist groups using drones, but from this day forward, we must recognize that drone use by terrorist-designated groups is not just a future threat. It's real, and it's happening every day at our borders."[1] Many of the designated cartels and TCOs have long used UAS in Mexico and near U.S. borders for a variety of purposes.

For more than a decade, the use of UAS has increased the operational capabilities and tactical impacts of terrorist actors around the world, including affiliates of the Islamic State, al-Qaeda and the Iran-backed Axis of Resistance. The threat is not new, but it is

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evolving and remains unresolved. Notable advancements in UAS-related technology mean that drone payloads, flight times and sensor capabilities are improving significantly and at a rapid pace—and become more accessible to a broader base of lay users.

There is concern that cartels may use UAS to attack U.S. security forces. For instance, according to new reporting, a February 2025 law enforcement memo advised that "Mexican cartel leaders have authorized the deployment of drones equipped with explosives to be used against US Border Patrol agents and US military personal currently working along the border with Mexico." In April 2025, Air Force Gen. Gregory M. Guillot—commander of U.S. Northern Command (NORTHCOM) and the joint U.S.-Canadian North American Aerospace Defense Command (NORAD)—testified to Congress that he "proposed…a change to the rule of force." He argues this would "allow us to shoot down or bring down drones that are surveilling over our deployed and mobile troops…not just that are in self-defense, but anything that's surveilling and planning the next attack on us within five miles of the border."

In short: the terrorist drone problem isn't going away—and, from the perspective of U.S. homeland security and defense—it is taking new form. The inclusion of FTO-designated TCOs actors raises a critical research question: *How does the adoption and use of drones by cartels and gangs differ from that of traditional terrorist groups—and why?*

This paper argues that terrorist organizations and cartels are converging technologically while diverging strategically. Both exploit commercially available drones for surveillance, intimidation and explosive delivery, but they do so in ways shaped by different objectives, resources and support networks. Iran-backed terrorist organizations have advanced furthest along the adoption curve because state sponsorship provides access to military-grade systems, training and logistical support. By contrast, terrorist groups without state sponsorship and cartels rely more heavily on commercial platforms, improvised modifications and localized tactical adaptation.

This convergence has important policy implications. UAS proliferation is eroding traditional distinctions between terrorism, insurgency and organized crime. Cartel drone use increasingly resembles paramilitary or insurgent behavior, while terrorist drone use often depends on the same commercial supply chains and improvised modifications available to transnational criminal actors. These patterns suggest that counter-drone policy should move beyond separate counterterrorism and counternarcotics frameworks and instead address the shared technological ecosystems through which violent non-state actors acquire, adapt and operationalize drones.

Several developments have accelerated this convergence in drone use across violent non-state actors. Continued improvements in commercially available drone technology have reduced barriers to acquisition by increasing range, payload capacity, autonomy and ease of operation while lowering costs. At the same time, increasingly interconnected transnational networks have facilitated the diffusion of tactical knowledge through online communities, battlefield experience and organizational linkages. Competition among violent non-state actors has further encouraged rapid technological adaptation, as groups seek operational advantages over both rivals and state security forces. More recently, the conflict following the 7 October 2023 Hamas attack and the war in Ukraine have demonstrated the operational value of drones and accelerated learning and imitation across terrorist organizations and criminal networks.

To examine these dynamics, we compare how terrorist organizations and transnational criminal organizations adopt, adapt and operationalize unmanned aerial systems. We combine a conceptual framework with qualitative case studies and event-level analyses of ACLED and GRID data to identify similarities and differences in drone adoption, operational use, targeting patterns and lethality across actor types. We conclude by discussing the implications of these findings for counter-drone policy, homeland security and future research.

Contextualizing Terrorists and Transnational Criminal Organizations

The inclusion of the transnational gangs and drug cartels to the list of Foreign Terrorist Organizations brought together a set of actors that have traditionally been associated with distinct threat streams and, as a result, are often considered the responsibility of distinct government counter-threat units with unique sets of authorities, tools and equities. As a foundation of our argument and empirical analysis, we briefly summarize what the existing academic and grey literatures[2] tell us about the general objectives, structures and motivations of traditional terrorist organizations and those of cartels and transnational criminal organizations. We are certainly not the first to consider these actor groups in comparison. A substantial body of scholarship has explored this comparison and the broader crime-terror nexus.[3]

Terrorist Organizations and Networks

The 2002 Homeland Security Act defines the act of terrorism as "any activity that involves an act that is dangerous to human life or potentially destructive of critical infrastructure or key resources; and is a violation of the criminal laws of the United States or of any State or other subdivision of the United States; and appears to be intended to intimidate or coerce a civilian population; to influence the policy of a government by intimidation or coercion; or to affect the conduct of a government by mass destruction, assassination, or kidnapping."[4] This definition speaks to the observed nature, purpose and common means and targets that can help separate terrorism from other violent criminal acts. As an action, terrorism is a means to an end, a method by which non-state actors threaten or wield illegal violence with the ultimate aim of affecting change and advancing an ideological cause or political aim.

Individuals, networks and organizations may advocate and use terrorism as a consistent means of achieving their goals. In some cases, these are formally designated as terrorist entities. For example, the U.S. Secretary of State designates Foreign Terrorist Organizations in accordance with Section 219 of the Immigration and Nationality Act and based on the following legal criteria: the organization must be a foreign organization; the organization engages in terrorist activity or terrorism or retains the capability and intent to engage in terrorist activity or terrorism; and the terrorist activity or terrorism of the organization threatens the security of United States nationals or the national security of the United States.[5] The current list of FTOs designated by the United States includes the Islamic State and its global *wilayat* branches, al-Qaeda and many of its affiliated groups, Hamas, Hezbollah, Ansarallah (aka the Houthis), the National Liberation Army, the Haqqani network and others. Many of these organizations

and networks are behind armed insurgency campaigns against local regimes and security forces. Their main aim is gaining political authority and control over a defined political territory—whether *via* regime takeover, secession, or otherwise. Terrorism is one of several forms of organized violence that they employ to achieve those goals.

Cartels and Gangs

Transnational criminal organizations are self-perpetuating associations of individuals who operate transnationally for the purpose of obtaining power, influence, monetary and/or commercial gains, wholly or in part by illegal means, while protecting their activities through a pattern of corruption and/or violence, or while protecting their illegal activities through a transnational organizational structure and the exploitation of transnational commerce or communication mechanisms. The United Nations Convention Against Transnational Organized Crime (UNTOC) defines an organized criminal group as a "structured group of three or more persons, existing for a period of time and acting in concert with the aim of committing one or more serious crimes or offences… in order to obtain, directly or indirectly, a financial or other material benefit."[6]

As organizations and networks, Mexican drug cartels are leading suppliers of illicit narcotics to the United States and other global markets. The 2024 U.S. Drug Enforcement Administration (DEA) *National Drug Threat Assessment* summarizes this succinctly,

> The Sinaloa and Jalisco cartels are called "transnational criminal organizations" because they are not just drug manufacturers and traffickers; they are organized crime groups, involved in arms trafficking, money laundering, migrant smuggling, sex trafficking, bribery, extortion, and a host of other crimes—and have a global reach extending into strategic transportation zones and profitable drug markets in Europe, Africa, Asia, and Oceania.[7]

Having large military forces, providing high levels of employment and controlling large annual revenues, cartels are deeply entrenched in the state regime and local political infrastructure. Levels of reported corruption, co-option and coercion across political institutions and officials are systemic. As Hazen notes, while the Mexican drug cartels aim to intimidate or infiltrate the government, they have no interest in driving state collapse, as such instability would be detrimental to their business.[8]

In practice, cartels and other TCOs employ violence in ways that are qualitatively similar to violent acts of terrorism as they "systematically target several government institutions to promote instability and influence their policy."[9] Moreover, intra-cartel violence and civilian targeting in Mexico contribute to tens of thousands of civilian deaths in the country each year. Assassinations of journalists and public servants are relatively common. Elections are commonly targeted to discourage civic participation and manipulate outcomes at national, regional and local levels. The 2024 election was especially violent, for instance—more than thirty candidates killed in incidents attributed to cartel-linked violence.

Cartels and Terrorists in Comparison

Historically, the academic literature has considered "methods and motives" as lines of separation marking the terror and transnational crime spaces. Phil Williams, for

instance, argues that at the core of terrorist organizations is the desire to bring about political change, while criminal organizations focus on profit generation and maximization. Of course, there are limitations to this framing. As Shelley and Picarelli point out: "Terrorists and criminals often use the same methods, most often for divergent motives—but not always."[10] Indeed, the authors find evidence of meaningful "points of convergence" between organized criminals and terrorists that are worth taking into account. In particular, they identify points of similarity in the behavioral and operational methods that both terrorists and TCOs have developed to Similarly, Longmire and Longmire conclude, "The tactics, strategy, organization, and (to a limited extent) the goals of the Mexican drug cartels are all perfectly consistent with those of recognized terrorist organizations."[11] Jenkins puts this well, arguing that cartel violence "serves a political agenda aimed at increasing their influence over policy, expanding their control over territory and state institutions, and deterring interference in their operations. The drug cartels directly threaten state authority as much as urban guerrillas."[12]

In short, while the strategic motivations of terrorist organizations and transnational criminal organizations remain distinct, advances in drone technology are producing increasing overlap in their operational capabilities and tactical behavior. This distinction between strategic divergence and technological convergence forms the central analytical framework of this study and guides the comparative assessment presented in the following sections.

Literature Review: Why Violent Non-State Actors Innovate or Adopt New Technologies

Over the past ten years, the academic literature has investigated the underlying logic by which violent armed non-state actors—whether insurgent terrorist actors like the Islamic State or transnational criminal organizations like the Sinaloa Cartel—innovate or adopt new technologies and related tactics. In many cases, this includes decisions and cycles of innovation, success and failure, establishing mechanism of acquisition and development and tactical adaptation. These studies explore a range of applications from the diffusion of suicide bombing tactics to the use of cryptocurrencies to the use of additive manufacturing to the malign exploitation of extended reality. We summarize this literature as well as advancements made by academic studies related to violent non-state actor drone operations.

Technological Innovation and Adoption

At its core, innovation focuses on solving a problem with the implementation of an idea that is both novel and useful.[13] In a malevolent context, threat actors are innovative when they adopt ideas that are novel and useful and generate surprises. Militant organizations and networks provide a unique context for examining tactical innovation, which often serves as a crucial mechanism for survival and advancement.[14] While the life span of most terrorist organizations is notably short—many disband or are dismantled within a year of formation—those that persist tend to do so by continuously adapting and innovating in response to the shifting demands

of their operational environment.[15] Similar to other organizational types, violent non-state actors leverage creative strategies and tactics to enhance their effectiveness and to outmaneuver both competing factions and state security forces. At the tactical level, innovation can return a variety of desired outcomes. It can "confer combat advantages (on the battlefield or within a contested operational space) as well as organizational benefits (including enhanced prowess and strategic reach."[16] Successful innovation may also boost the reputation of armed militant actors, enabling them to project strength, legitimacy and influence within their constituencies.[17]

Experts from a number of academic fields—including security studies and military history, terrorism, industrial-organizational psychology, political science and others—explain how and why some organizations choose to innovate their tactics or adopt novel methods, specifically through the adaptation of technology. Davis argues that VNSAs decide to adopt new technology based on perceptions of its usefulness and of its ease of use.[18] While usefulness aligns with the expected benefit—whether at the strategic, operational, or tactical levels—the perceived ease of use refers to accessibility, low cost of entry and ease of operation. Cronin advances this argument finding that terrorists adopt new technologies most readily when innovations emerge from "open" technological revolutions—i.e. those that are widely accessible, commercially available and not tightly controlled by states.[19] Rather than inventing cutting-edge weapons themselves, violent non-state actors tend to co-opt technologies originally developed for civilian or benign purposes, taking advantage of the broad diffusion of knowledge and tools in a defined open innovation era.

Once decisions are made, certain organizations are better at implementation than others. In a militant context, these related processes—innovation and adoption through diffusion—are shaped by several factors, including the organization's leadership, network ties, available resources, recruitment practices and environmental conditions. It is in this context that we approach our investigation into the logic of terrorist and cartel adoption and use of drones.[20]

Violent Non-State Actor Use of Unmanned Aerial Systems

Terrorist groups and cartels share common operational and tactical needs and for both, drones serve as a novel form of air power.[21] Drones provide these actors with new means of coercion through attrition, reputational enhancement and intimidation of local populations.[22] They also support a wide range of tactical applications, including pre-attack planning, intelligence, surveillance and reconnaissance (ISR), kinetic strikes, logistics and post-attack exploitation. As force multipliers, drones enable greater precision, expand operational reach and enhance the disruptive effects of violent campaigns.

Rassler and Veilleux-Lepage summarize the first major wave of militant drone operations.

> During the mid-2010s, groups such as the Islamic State and al-Qàida rapidly weaponized commercially available drones, employing them for surveillance, bomb delivery, propaganda, and psychological operations. These developments—the sudden introduction of new capabilities that transformed operational practices—represent the "short bursts of rapid

change" that disrupted the existing status quo or equilibrium. Following this wave of innovation, a new equilibrium emerged, as many terror groups refined their drone strategies, adopting methods similar to those of the Islamic State and al-Qàida, while others lagged due to resource constraints.[23]

More recently, the use of drones during the October 7th attack by Hamas were a central element in a "combined armed use", for intelligence, surveillance, reconnaissance (ISR) and for paralyzing Israel's command and control structure and degrading Israel's ability to respond in early stages by targeting precision strikes on Israeli observation and communication towers and coordinated suppression of defenses in concert with other tactics including, rockets and missiles.[24] This multidomain, synchronized assault, where drones facilitated the breach of hardened defenses, allowed Hamas to inflict outsized damage on October 7.[25] In addition to the highly public nature of the October 7 attacks, the ongoing conflict in Ukraine has proved a laboratory for drone innovation and an inspiration for armed groups around the world. For example, Milnes and Lyammouri recently found that "the low-cost, field-adapted model of drone warfare seen in Ukraine appears to have influenced how armed groups in the Sahel are approaching drone integration."[26]

As an additional point of value, the current cost-ratio stills favors militants who get outsized bang for their buck while requiring security forces to expend considerable resources to mitigate the drone threat. Employed at scale, this can coerce security forces to allocate and prioritize resources in ways they might not, absent the use of drones. Commercial and even custom homemade drones are relatively inexpensive compared to the advanced and costly counter-drone technologies used by state actors. Even when drones are less lethal than other weapon system options (eg VBIEDs), they have been effective in generating significant economic harms as well as fear on the targeted population. Multi-drone attacks—including swarms—fit into this framework by enabling terrorists to bypass traditional defenses and strike targets deep within adversary territory, much like traditional air strikes. Even without mass casualties, drones can create a persistent threat environment, without risking operatives, preserving terrorist militants and sustaining campaigns over time.[27]

As Kerry Chávez and Ori Swed have argued elsewhere, the use of drones can also give militants a reputational boost among supporters and potential recruits as they use these to "advertise their newfound aerial capabilities and their effects and to publish striking cinematography of other operational successes."[28] The acquisition and operational deployment of drones showcases a group's technological sophistication and supports outbidding efforts—signaling superior capability and commitment compared to rivals, such as in the competition between al-Qaeda and ISIS affiliates in Africa to develop drone capacity. Drones also facilitate intimidation by generating fear and insecurity among civilian populations, even from remote areas.

Although FTO-designated terrorist organizations and cartels glean similar operational and tactical benefits from the use of UAS, differences in their core motivations and logistical and social constraints are expected to lead to variation in the scale and sophistication of their drone use. Terrorist organizations are, by definition, politically motivated actors seeking outcomes such as regime change, territorial control, policy change, social control, or the status quo maintenance, they pursue these goals through strategies such as attrition, intimidation, provocation, spoiling and outbidding.[29] For

high-resource drug cartels, innovation—and its application at scale—are determined to be useful when it advances broader efforts to gain material gains and territorial control over places and criminal enclaves, establish market domination and increase intimidation and political influence.[30] They are primarily motivated by the need to sustain their illicit operations and protect their business and operations from the intervention of government and rival groups.[31]

Ultimately, the relative benefits of adopting and deploying drones will be strongly shaped by an organization's strategic objectives, resources, operational preferences and tactical capabilities. As such, there are reasons to expect that the timing, nature and scale of TCO drone operations may share both points of overlap and distinction from those of traditional FTOs. We explore this in a structured assessment in the following section.

Understanding Adoption and Deployment of Unmanned Aerial Systems

Although violent non-state actors are increasingly experimenting with a broader range of unmanned systems, including unmanned ground vehicles (UGVs) and unmanned surface and underwater vehicles (USVs/UUVs), this study focuses exclusively on unmanned aerial systems (UAS). We adopt this narrower scope because aerial platforms remain the most operationally mature, operationally significant and extensively documented unmanned technology employed by terrorist organizations and transnational criminal organizations.

Both traditional foreign terrorist organizations (FTOs) and transnational criminal organizations (TCOs) have adopted UAS across a range of operational roles. They employ UAS throughout the attack cycle, including surveillance, target selection, weapons acquisition and employment, logistics, attack and post-attack exploitation. UAS platforms can be tailored and configured for these distinct tasks.[32]

For non-state violent actors, innovation is not optional; it is a necessity for survival. Those that stagnate may quickly lose relevance and power. Groups that fail to evolve risk being dismantled by state actors or outpaced by competing organizations. The pressure to stay ahead is intensified by competition over limited resources, including funding, recruits and public support. Those unable to keep pace with this technological arms race quickly lose relevance and power.[33]

In addition to their operational utility, how groups adopt and use drones are profoundly shaped by both the actual and perceived ease of access, dimensions closely linked to the financial cost of entry and the capacity to acquire, construct, or modify such systems. In terms of usability, commercial off-the-shelf (COTS) drones are particularly salient owing to their intuitive, game- or smartphone-like control interfaces, which markedly diminish training demands and enable operators with limited or no prior experience to employ them effectively.[34] Moreover, beyond their widespread commercial availability, many COTS platforms exhibit a high degree of adaptability, permitting the integration of payload-release mechanisms, optical sensors and other modular components with minimal technical expertise. This inherent modifiability simultaneously facilitates the repurposing of these technologies by malicious actors for a range of illicit applications, including reconnaissance, smuggling and the delivery of explosive or otherwise hazardous materials.[35]

To characterize patterns of technological adoption, we draw on the Violent Non-State Actor (VNSA) Technology Adoption Curve proposed by Gartenstein-Ross, Clarke and Shear,[36] which conceptualizes innovation as progressing through four broad stages: early adoption, iteration, breakthrough and competition. We use the model as a comparative heuristic for examining how different organizations adopt and operationalize UAS. Our objective is not to validate the framework itself, but to assess how organizational objectives, external support and operational environments shape progression through each stage.

Our application of the framework requires two important clarifications. First, progression through the stages should not be interpreted as a single evolutionary pathway shared by all violent non-state actors. Some organizations adopted drone technology through documented knowledge diffusion—for example, among ISIS affiliates or Iran-backed proxy networks—whereas others independently developed similar capabilities in response to comparable operational challenges. Second, although this study compares terrorist organizations and transnational criminal organizations, these categories are used as analytical groupings rather than rigid taxonomies. Some organizations exhibit hybrid characteristics or facilitate cross-organizational diffusion of tactics and expertise. Accordingly, we use representative cases to illustrate broader patterns of technological adoption rather than to imply direct learning relationships among every organization discussed.

As seen in Table 1, we operationalize this framework by defining the criteria used to distinguish each adoption stage and providing representative examples illustrating how terrorist organizations and transnational criminal organizations progressed through them.

Table 1. Operational criteria and comparative progression through the adoption curve.

Stage	Operational stage	Illustrative terrorist examples	Illustrative cartel/TCO examples
Early adoption	Initial experimentation; isolated incidents; limited operational effectiveness; little or no sustained organizational integration.	Aum Shinrikyo's RC helicopter experiment; Virginia Jihad Network; FARC radio-controlled aircraft; Hezbollah reconnaissance flights.	Early commercial drone use for ISR and smuggling (early 2010s); initial cross-border drug transport; limited operational sophistication.
Iteration	Repeated operational use accompanied by organizational learning, technological refinement and expansion into multiple mission sets (eg ISR, propaganda, weaponization).	ISIS develops formal drone program, standard operating procedures and weaponized quadcopters; Hezbollah refines ISR and attack capabilities.	Cartels systematically employ drones for surveillance, border monitoring, smuggling and early weaponization (eg CJNG "Papa Bomba"); incorporation of lessons from external actors.
Break-through	Drones become routinely integrated into offensive operations and serve as force multipliers in coordinated campaigns; sustained operational capability demonstrated.	Houthis, Hamas, Hezbollah, ISWAP and JNIM integrate drones into combined-arms operations, precision strikes and long-range attacks.	CJNG and LNFM routinely employ explosive drones for territorial control, attacks on rivals and security forces and coordinated paramilitary operations.
Competition	Widespread deployment of counter-UAS measures prompts adaptation through new tactics or technologies designed to overcome defensive systems.	Iran-backed groups adapt to layered air defenses; growing use of FPV and fiber-optic technologies; continued adaptation following counter-UAS advances.	Emerging adoption of counter-counter-UAS tactics, including jammers, commercial counter-UAS equipment and reported experimentation with fiber-optic-controlled drones

Terrorist Groups

Early adoption by terrorist groups spans from 1990s to the late 2000s, marked by crude experimentation, high failure rates and little to no sustained operational impact. These examples illustrate independent experimentation rather than documented organizational learning. Groups typically relied on early civilian model aircraft, fixed-wing, radio-controlled platforms without cameras. These systems lacked accuracy, payload capacity and range, making them incapable of delivering precise, stand-off and destructive effects.[37] Most attempts were disrupted by state authorities or ending in technical failure.[38] Gartenstein-Ross emphasizes that failures during these phases do not necessarily hinder these groups; rather, as they note, "Botched terrorist attacks aren't failures for terrorist groups. They're a learning process."[39] This phase was particularly advantageous for groups. Since state actors have not yet developed effective defensive policies, legal frameworks, or counter-technologies, VNSAs often operate with minimal interference from law enforcement or intelligence agencies.

A key historical example of early unmanned aerial vehicle experimentation comes from the Japanese doomsday cult Aum Shinrikyo in 1994. The group attempted to use a remote-controlled Mi-17 helicopter to disperse a large amount of sarin, a highly lethal nerve agent to murder the leader of a rival spiritual organization. However, the experiment failed, with the helicopter crashing on their second attempt.[40] Although these early cases occurred in different organizational and regional contexts, they collectively illustrate independent experimentation with unmanned technologies rather than documented organizational learning.

In the context of North America, Gartenstein-Ross and his colleagues report that the earliest known instance of a terrorist group attempting to employ drones for malicious purposes dates back to 2002.[41] The group, later identified as the Virginia Jihad Network, sought to acquire drone components and related technologies to support Lashkar-e-Taiba (LeT), a Pakistan-based terrorist organization. As part of these efforts, the group successfully purchased a system capable of autonomously flying a drone to predetermined GPS coordinates; however, the equipment ultimately failed to function as intended. By 2003, U.S. authorities had arrested eleven members of the Virginia Jihad Network.[42] Around the same time, in 2002, Colombian authorities raided a FARC camp and discovered nine radio-controlled airplanes, one already loaded with explosives. The find revealed an intent to weaponize hobbyist aircraft for aerial delivery, but with limited capacity: payloads under two pounds, limited endurance and short operational ranges.[43] Concurrently, non-state actors in the Middle East also began experimenting with drones. The first reported use of drone by a non-state actor against a state goes back to November 2004, when Hezbollah deployed an Iranian-made Mirsad-1 drone on a reconnaissance mission over Israeli territory. The drone flew undetected for about 20 min before returning to Lebanon.[44]

During this early adoption phase, the technology showed promise but was constrained by decisive limitations, including short range, minimal payload capacity and imprecise targeting. In addition to these technical shortcomings, the processes of acquisition and preparation frequently exposed militant groups to state intelligence, with most attempts failing to evade detection. Nevertheless, this trial-and-error stage served as a critical learning experience, fostering technical proficiency over time.

The iteration phase, occurring in the late 2000s through late 2010s, refers to the period when terrorist actors began refining their use of technology and integrating drones into broader operational strategies. Advancements in commercial technology facilitated this process by lowering barriers to entry and making drones easier to operate.[45] We argue that this phase spans from the early to the late 2010s.

Nearly a decade after the Virginia Jihad Network plot, in 2011, the FBI arrested Rezwan Ferdaus, a U.S. citizen linked to al-Qaeda, after uncovering his plan to attack the Pentagon and U.S. Capitol with model airplanes packed with explosives. The scheme illustrates how actors are not just learning whether the technology works, but also how to better integrate and optimize it to their advantage. The plotter planned to acquire remote-controlled drone-like aircraft, load them with explosives and direct them into the Pentagon using an onboard GPS guidance system.[46] Ferdaus's key innovation, though imperfect, was his use of a commercially available GPS autopilot system and Google Earth to navigate the aircraft toward their targets.[47] Incorporating the GPS guidance system, rather than manually flying the device within line of sight, marked a leap in user-friendliness. Further technological advancements in imaging and navigation systems have enabled violent non-state actors to locate, track and target specific individuals and groups with increasing precision. For example, in September 2014, Hezbollah conducted one of the first known lethal drone strikes in Arsal, Syria, killing more than 20 fighters from the al-Nusra Front.[48]

Another early indicator of this shift emerged in the Middle East, when the Islamic State (ISIS)[49] began using drones to capture aerial surveillance footage for propaganda videos in mid-2014.[50] By 2015, ISIS had established a formal program and standardized procedures for drone operations. This bureaucratic system, complete with mission forms, checklists, post-mission reports and analyses of both failed and successful flights, significantly contributed to the organization's adoption of drone technology.[51] Drones have become force multipliers and being used as combined weapons to guide vehicle-borne suicide bombers through urban streets, adjust mortar and rocket fire for greater accuracy and coordinate suicide attacks. For instance, during the defense of Mosul in 2016 and 2017, ISIS drone operators acted as navigators, directing suicide drivers in real time through battle-damaged streets.[52] ISIS's first recorded use of a weaponized drone occurred in 2016, when a device recovered by Kurdish Peshmerga forces exploded during inspection on October 2, killing two fighters and injuring two French soldiers.[53] Three months later, in January 2017, ISIS released a propaganda video showing modified commercial quadcopters dropping munitions with notable accuracy.[54]

As Rassler reports, "many of the commercial drones that the group acquired were then creatively modified through the cobbling together of cheap, easily acquired add-on components that allowed the group to moderately enhance the capabilities of those drones."[55] Acquisition documents obtained from the battlefield show that ISIS's drones were not highly sophisticated but relied on commercially available parts such as GoPro cameras, GPS units and propeller blades. At the same time, the group demonstrated a resourceful DIY capacity, seeking encrypted video transmitters, long-distance radio relay devices and servo motors to extend range and enhance performance, which can be also used to release mechanisms for munitions.[56] These weaponized drones have been described as devices "a sophisticated high schooler could put together".[57]

In Africa, drone adoption occurred later, with the early indications of jihadist interest emerging in 2014, when members of the Islamic State West Africa Province (ISWAP) sought technical guidance from Syrian jihadists to repurpose civilian drones for militant use in the Lake Chad region, reflecting the diffusion of battlefield innovation across transnational networks.[58] The earliest verified use of weaponized drones by African Islamist groups came in 2018, when Boko Haram employed drones for intelligence, surveillance and reconnaissance (ISR) on military personnel.[59] As Figueiredo notes, African affiliates of ISIS and al-Qaida benefited from manuals, online tutorials and training provided by fighters returning from the Middle East.[60] By 2018, ISWAP had modified commercial quadcopters to deliver improvised explosive devices against Nigerian military positions, shifting from reconnaissance to offensive use. Similar patterns emerged among Jama'at Nasr al-Islam wal Muslimin (JNIM) in the Sahel, which employed drones for surveillance and harassment of French and Malian forces and al-Shabaab in Somalia, which used them for both propaganda and attacks.[61]

By the late 2010s, terrorist organizations had moved beyond trial-and-error experimentation. Iran-backed groups and ISIS and al-Qaida affiliates had achieved notable progress in integrating drones across propaganda, reconnaissance and combat operations.

The breakthrough phase, occurring from the late 2010s through the mid 2020s, marks a pivotal stage in which terrorist actors transitioned from the experimental use of drones to their consistent and operationally reliable deployment. According to Gartenstein-Ross et al., the release of the DJI Phantom 1 in 2013 marked a key milestone in the transition to the breakthrough stage of drone adoption.[62] The Phantom 1 revolutionized the consumer drone market by introducing affordable, user-friendly drones equipped with cameras, GPS and autonomous flight capabilities. However, we contend that the true breakthrough for terrorist organizations occurred in the late 2010s and early 2020s, driven by the increasing accessibility and enhanced performance of commercial off-the-shelf (COTS) drones and further accelerated by innovations emerging from the war in Ukraine.

In their analysis of 22 DJI drone models released between 2013 and 2024, Dulligan and colleagues presented a consistent trajectory of performance improvements, including velocity, transmission distance and flight duration of DJI drones over time.[63] These advancements have enabled operators to deploy drones from greater distances and sustained longer missions. In addition, the price barrier has declined, while performance has steadily increased.[64] This trend is largely attributable to the broader reduction in component costs, driven by the widespread availability of electronic sensors, GPS devices and Wi-Fi receivers, which has provided affordable and flexible opportunities for users to assemble and customize drones for their own purposes.[65] Concurrently, advancements in drone autonomy and avionics, particularly in obstacle avoidance and vertical takeoff and landing (VTOL) technologies, have made civilian drones increasingly intuitive, streamlined and easy to operate.[66]

This evolution has been further amplified by the proliferation of military-grade drones among Iran-backed groups in the Middle East. These systems provide extended range and increased payload capacity, enabling more sophisticated and far-reaching operations. As a result, such groups have used drones not only to strike domestic targets in Yemen, but also to project power beyond their borders, targeting critical infrastructure in Saudi Arabia and the United Arab Emirates, as well as vessels in the

Red Sea and the Gulf of Aden and even Israel, located over a thousand miles from Houthi-controlled territory.[67] The 7 October 2023, Hamas attack on Israel triggered a sharp escalation in regional conflict and Iran-backed groups, including Hizballah, the Houthis and the Islamic Resistance in Iraq, intensified their attacks on Israeli and U.S. assets across the region, particularly through drones and missiles.[68]

The Al-Qeada and ISIS affiliated groups in Africa reached its breakthrough phase in drone warfare several years after its Middle Eastern counterparts. By late 2024, the Islamic State West Africa Province (ISWAP) had integrated drones into coordinated attacks on Nigerian military formations, inflicting casualties and disrupting operations.[69] Unlike earlier isolated incidents, these operations demonstrated a sustained capacity to incorporate drones into complex, multi-phase offensives. Similarly, Al-Shabaab advanced its operational sophistication by employing drones for surveillance of AMISOM bases and later using them to support high-profile attacks in Somalia. In northern Mozambique, ISIS-linked insurgents began deploying drones to monitor Mozambican and allied forces.[70]

According to a 2025 United Nations Security Council (UNSC) report, "As at late 2024, Member States reported that 13 ISIL (Da'esh) trainers had arrived in the Lake Chad Basin from the Middle East, facilitating the acquisition, assembly and deployment of drones…".[71] Another accelerating factor has been non-state actors' exposure to the Ukrainian conflict, particularly that of Jama'a Nusrat ul-Islam wa al-Muslimin (JNIM), where groups, including jihadist organizations and criminal cartels, have sought first-person-view (FPV) drone training. The UNSC report further highlights that Al-Qaeda (AQ) and ISIS-affiliated groups, particularly JNIM and the Islamic State West Africa Province (ISWAP), have rapidly evolved from rudimentary drone experimentation to full battlefield integration.[72] By mid-2025, JNIM was routinely employing drones for reconnaissance and precision strikes across Mali, Burkina Faso and Togo.

As the breakthrough stage matured, drones became increasingly prevalent, shifting from the periphery to the center of contemporary conflicts. They have emerged as a core component of violent operations, functioning as force multipliers and enablers and have eroded the long-standing monopoly of states over the air domain.

The competition stage, upon which we may only now be entering, marks a critical turning point in which violent non-state actors face systematic and coordinated countermeasures from their adversaries. Since the onset of drone proliferation, state agencies and private industry have developed diverse methods to detect, identify and neutralize hostile drones. By 2018, the Center for the Study of the Drone at Bard College had identified over 150 companies producing counter-drone technologies.[73] Since then, counter-UAS (C-UAS) capabilities have evolved rapidly, moving beyond traditional radar and radio-frequency tools toward multi-sensor systems integrating LiDAR, acoustic, electro-optical/infrared and AI-driven detection and tracking. Modern C-UAS platforms combine soft-kill measures (eg GNSS and RF jamming or spoofing) with hard-kill solutions such as directed-energy weapons and kinetic interceptors. While initially pioneered by the military, these technologies are increasingly being adapted for use by law enforcement, border security and critical infrastructure protection agencies.[74]

The adoption of counter-drone technologies has been uneven across governmental and security agencies. Evidence from recent conflicts in Ukraine and the Middle East

demonstrates that effective defense requires far more sophisticated and costly systems than offense. Consequently, governments must secure reliable C-UAS supply chains and allocate substantial financial resources to harden vulnerable targets against drone threats.[75]

The U.S. Department of Defense (DoD) has invested heavily in strengthening against drone threats.[76] These sustained efforts proved critical following October 7, when Iran-backed groups sharply escalated drone attacks against U.S. forces, Israel and international shipping in the Red Sea and Gulf of Aden. Between October 2023 and October 2024, 373 UAS attacks were recorded, 153 targeting U.S. forces. Despite the surge in attacks, fatalities remained low, with the average death rate per attack falling by 93%, compared the previous year, which reflects the effectiveness of layered U.S. and Israeli air defense systems.[77] To strengthen the U.S. homeland's defense against drone threats, the White House issued the Restoring American Airspace Sovereignty Executive Order in June 2025. As a follow-on measure, the U.S. Department of Defense established Joint Interagency Task Force 401 (JIATF 401) in August 2025, consolidating expertise, resources and operational capabilities across military branches and federal agencies under a centralized command structure.[78]

While counter-drone initiatives have become increasingly capable and widespread, adversaries continue to adapt, perpetuating what Gartenstein-Ross et al. describe as a "cycle of adaptation and counteradaptation."[79] Drawing lessons from the ongoing war in Ukraine and Russia, these actors have developed innovative evasion tactics, for example, the use of fiber-optic links to bypass jamming. Notably, the Azawad Liberation Front (FLA) in Mali reportedly became the first non-state actor to employ fiber-optic FPV drones, targeting Malian government forces and the Russian African Corps.[80]

Cartels and Other Transnational Criminal Organizations

The early adoption of drones by drug cartels dates back to the early 2010s, significantly later than their use by terrorist organizations.[81] Early on, cartels repurposed commercial and recreational drones for intelligence, surveillance and reconnaissance (ISR) and smuggling.[82] By 2015, narco-drone incidents began attracting media coverage, such as when a drone carrying over six pounds of methamphetamine crashed near Tijuana, just 15 miles from San Diego and when U.S. Customs and Border Protection intercepted another drone in Calexico loaded with nearly 29 pounds of heroin.[83]

Compared to terrorist groups' early experimentation with small model aircraft, cartels began employing drones only after the technology had advanced to quadcopters equipped with first-person-view (FPV) systems and enhanced maneuverability, making them suitable for intelligence, surveillance and reconnaissance (ISR), smuggling and eventually precision strikes.[84] However, these initial efforts were limited, as cartels relied on commercial off-the-shelf (COTS) drones with restricted payload capacity and short operational range at the time.[85]

Between 2012 and 2014, the U.S. Drug Enforcement Administration recorded approximately 150 confirmed drone incursions related to cross-border narcotics trafficking, followed by over 170 additional incidents between 2015 and 2020.[86] In addition to smuggling, cartels increasingly employed drones for intelligence, surveillance and reconnaissance (ISR).

The iteration phase for Mexican drug cartels evolved from mere experimentation to strategic and systematic applications, driven by the need to sustain and protect their illicit operations. Cartels adapted drone technology to monitor territory, track rivals and law enforcement, streamline the cross-border distribution of drugs and weapons and conduct direct drug deliveries to customers, thereby reducing operational risks.[87]

The iterative phase of cartel drone adoption is not solely an internal process of trial and error; rather, it involves actively learning strategies from a wider landscape of global conflict and groups, such as Hezbollah.[88] The flow of tactical knowledge and inspiration between these groups has accelerated the learning process for cartels.

By the mid-2010s, cartels began experimenting with weaponization. The first publicly documented case occurred in 2017, when Mexican Federal Police in Guanajuato intercepted a 3DR Solo quadcopter armed with an improvised explosive device, attributed to the Cártel de Jalisco Nueva Generación (CJNG), an incident dubbed the "Papa Bomba."[89] In 2018, drones equipped with grenades targeted the residence of Baja California's public safety secretary in Tecate.[90]

Cartel adoption of drones did not occur in isolation. U.S. intelligence reports indicate that Mexican operatives received training from the Revolutionary Armed Forces of Colombia (FARC), while Hezbollah networks facilitated the transfer of arms and tactical expertise into Mexico.[91] These examples should not be interpreted as suggesting a single pathway of knowledge transfer between terrorist organizations and cartels. Rather, they illustrate that cartels have drawn upon multiple external sources of tactical innovation while simultaneously adapting commercially available technologies to their own operational environments. Although terrorist groups experimented earlier with rudimentary drone models, cartels entered the early adoption stage later on, but quickly exploited advanced commercial quadcopters for intelligence, surveillance, reconnaissance (ISR), smuggling and weaponization.

In April 2019, when U.S. Border Patrol agents working El Paso sector, they observed a drone functioning as a "lookout" device, providing intelligence on law enforcement activity and helping smugglers coordinate safe entry points for migrants.[92] Since this initial case in 2019, law enforcement agencies have reported a significant increase in drone-assisted smuggling operations along the border.

Taken together, these incidents reflect a clear trajectory of tactical evolution. What began as basic surveillance, reconnaissance and early experimentation with drone weaponization gradually advanced as groups refined their methods and expanded their technological acquisition, leading to successful drone operations.

The breakthrough stage for Mexican cartels coincided with their growing adoption of paramilitary tactics. Armed with increasingly military-grade weapons and equipment, groups such as the Cártel de Jalisco Nueva Generación (CJNG) now operate as fully equipped paramilitary forces.[93] Their arsenals extend well beyond conventional firearms, including handguns and automatic rifles, to include weaponized drones carrying explosives and chemical agents, as well as armored vehicles outfitted with heavy machine guns, rocket-propelled grenades, improvised mortars and land mines.[94]

According to the Armed Conflict Location & Event Data Project, the Cártel de Jalisco Nueva Generación (CJNG) deployed explosive-laden drones in Tepalcatepec in May 2021.[95] A few months later, in August 2021, CJNG launched additional multi-day drone assaults in the same region. On 10 January 2022, in Bejuco, Michoacán de

Ocampo, CJNG conducted a drone attack that dropped explosives on residents, injuring one person. Three days later, on January 13, another drone strike targeted Tepalcatepec, damaging several homes. These cases illustrate CJNG's transition from using drones for smuggling to employing them in targeted attacks against law enforcement and rival groups.

Data reported by the Secretariat of National Defense (SEDENA) of Mexico, drone attacks surged significantly over recent years: from only five incidents in 2020 to 107 in 2021 and then more than doubling to 233 in 2022. In the first six months of 2023 alone, SEDENA documented 260 such attacks.[96] During the same period, drone activity along the U.S. border also surged; authorities in the Rio Grande Valley reported over 10,000 cross-border incursions and 25,000 detections in 2022.[97]

Drawing on battlefield experience and tactical lessons from abroad, cartels have begun to professionalize their drone operations and integrate advanced combat methods. Mexican and Colombian cartel operatives are reported to have received training in Ukraine's International Legion to acquire first-person-view (FPV) kamikaze drone tactics and countermeasures.[98] In Michoacán, Mexican authorities arrested more than 50 Colombian and Venezuelan nationals—some reportedly former soldiers—recruited to train cartel members.[99] Moreover, Lyammouri and Milnes note that Mexican cartels such as the Sinaloa organization have long leveraged Al-Qaeda–linked networks in West Africa to facilitate cocaine trafficking through coastal ports.[100] This overlapping infrastructure not only supports narcotics smuggling but also enables potential exchanges of tactical expertise, including drone warfare techniques.

In late March 2024, La Nueva Familia Michoacana (LNFM) conducted a 72-h drone offensive in Petatlán, Guerrero, using explosive-equipped drones to ignite homes, livestock pens and other properties across multiple communities—displacing residents to secure territorial control. Over three days, approximately 40 bombs were dropped across several municipalities.[101] Later that year, on 16 December 2024, CJNG operatives launched explosive drone attacks against patrolling military personnel in Cotija de la Paz, Michoacán, killing two officers and injuring five others. This incident marked a significant escalation as it signals a shift toward deliberate strikes against state security forces.[102]

Beyond the use of explosive payloads, cartels have begun experimenting with toxic chemical irritants. Bunker documented four incidents between late 2022 and mid-2025 in which the Cártel de Jalisco Nueva Generación (CJNG) deployed drones carrying improvised chemical agents.[103] These included compounds such as aluminum phosphide, which is a toxic pesticide capable of causing hypoxia and circulatory failure.[104] The attacks targeted local residents and community police forces, marking a dangerous escalation that combined physical harm with psychological intimidation and demonstrated its ability to project power even into "safe" zones and to depopulate communities, forcing residents and civic leaders to flee.[105]

The **competition stage** for Mexican cartels has not yet fully emerged; however, the Trump administration's designation of major cartels as Foreign Terrorist Organizations (FTOs) in February 2025 is likely to accelerate their transition toward this stage.[106] Following the designation, the United States deployed military assets to the southern border, including radar and interdiction systems to support the Department of Homeland Security (DHS) and Border Patrol in counter-drone

operations. In Mexico, law enforcement agencies have also begun equipping units with counter–small unmanned aircraft systems (C-sUAS) to confront the expanding drone threat.[107]

However, Mexican cartels have demonstrated some adaptive resilience by integrating commercial counter-drone technologies, including jammers and detection systems, to shield their operations from both government surveillance and rival groups.[108] Despite incremental progress, official counter-drone initiatives, particularly in Mexico, remain underdeveloped and insufficient to meet the scale and sophistication of the threat.

Mirroring the adaptive behavior of terrorist organizations, cartels have begun modifying their tactics to neutralize C-UAS measures. According to Bunker, cartels have recently experimented with fiber-optic–controlled drones, importing the necessary components from abroad in direct emulation of Russian and Ukrainian battlefield technologies.[109] Operating through tethered fiber-optic links, these drones maintain continuous control in radio-frequency (RF)–denied environments, effectively circumventing jamming and spoofing. The deployment of such systems enables cartels to bypass U.S. and Mexican C-UAS defenses, particularly along the border, where RF and Global Navigation Satellite Systems (GNSS)-based interdiction remains the primary protective measure.

These adaptations compel governments and security agencies to reassess existing countermeasure frameworks and to develop more proactive, multi-domain risk management and defense strategies.[110] Ultimately, this dynamic reflects an ongoing cat-and-mouse cycle of technological adaptation, in which state and non-state actors continually refine offensive and defensive capabilities in response to one another.

Analytic Comparison

For a systematic cross-comparison of drone use by terrorist organizations and cartels, we used data from the Armed Conflict Location & Event Data Project. We retrieved the unlagged ACLED dataset on 7 January 2026, which includes incidents reported through 6 January 2026. For this analysis, however, we limited the dataset to incidents occurring through 31 December 2025 and excluded all events that took place in 2026.

As ACLED does not explicitly classify actors as "terrorist organizations" or incidents as "terrorism," we used U.S. State Department Foreign Terrorist Organization (FTO) designations to identify terrorist groups within the dataset. As FTO designation is not an academic classification and may be influenced by political considerations, we cross-validated these cases with the Global Terrorism Trends and Analysis Center (GTTAC) Record of Incident Database (GRID) dataset.[111] GRID applies the U.S. State Department's definition of terrorism and includes both designated and non-designated groups. As a robustness check on terrorist use of drones, we presented our analysis of GRID in Appendix A.[112]

Within the set of FTO-designated traditional terrorist groups in the ACLED, we distinguish Iran-backed organizations from other FTO-designated groups. For cartel-related incidents, we differentiated between FTO-designated cartels and other cartel/criminal groups. After filtering and classifying the groups, we had a comparison of temporal adoption trends, geographic clustering, lethality and civilian-targeting patterns associated with drone-involved violent incidents across groups.

Before presenting the results, it is important to emphasize that our objective is not simply to compare the frequency of drone incidents across actor types. Rather, we use these event data to examine whether terrorist organizations and cartels exhibit converging patterns of drone adoption while maintaining distinct operational behaviors. We therefore interpret the results through temporal adoption, geographic dispersion, civilian targeting and lethality.

Our analysis of ACLED (see Figure 1) reveals a clear divergence in the trends and persistence of drone adoption across actor types. Iran-backed terrorist organizations exhibited the most rapid escalation in drone use between 2020 and 2024, showing a sharp spike following 7 October 2023. This surge reflects the use of drones in coordinated regional campaigns involving long-range strikes, maritime attacks and combined-arms operations, the majority of which were directed against U.S. military installations in the Middle East and Israel.

Prior to October 7, the Houthis primarily targeted domestic sites in Yemen as well as infrastructure and military installations across Saudi Arabia and the United Arab

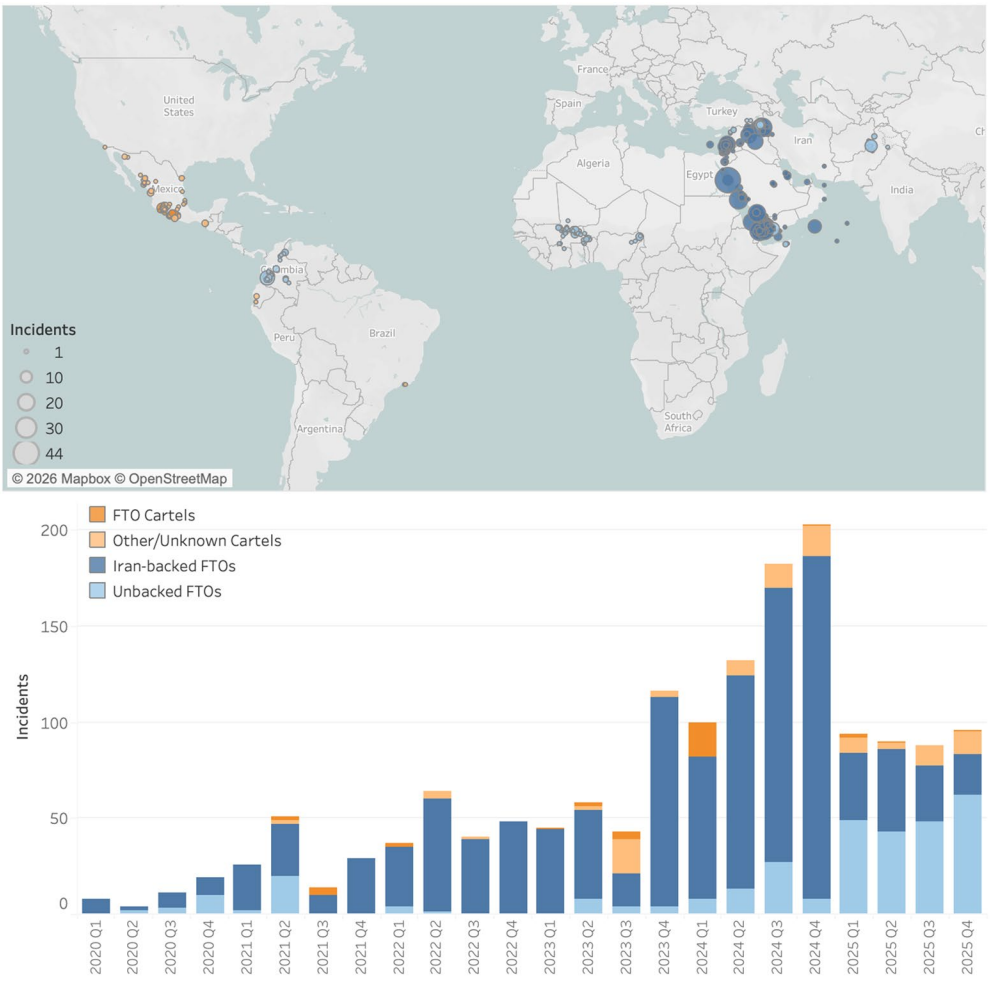


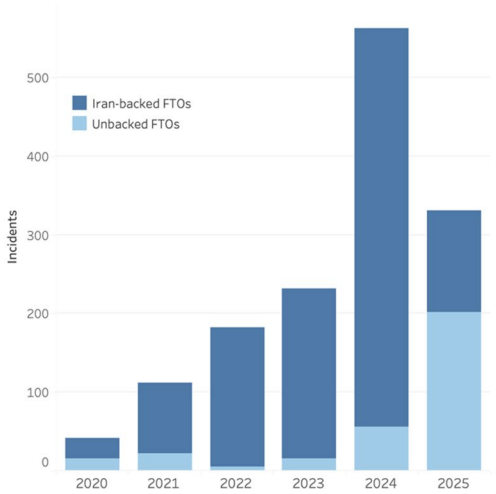
Figure 1. Drone attacks by terrorist groups and cartels, 2020–2025.[113]

Emirates. In the post–October 7 period, however, they expanded their operations to strike U.S. and Israeli positions and international commercial vessels in the Red Sea and the Gulf of Aden. State sponsorship has provided these groups, particularly the Houthis, with access to military-grade drones, including the Samad, Waid-2 and Qasef models, capable of carrying larger payloads and operating over extended ranges, thus enabling strikes thousands of miles from their base of operations. However, the data also indicate a notable decline in 2025, aligned with the effects of Israeli and U.S. operational attrition targeting Iran-backed proxy networks.

These temporal patterns demonstrate that convergence in drone adoption does not imply convergence in organizational capability. While all actor types have increasingly incorporated drones into their operations, the pace and sophistication of adoption differ substantially. State sponsorship enables terrorist organizations to accelerate beyond commercially available technologies, whereas cartels and non-state-sponsored terrorist groups continue to rely primarily on incremental adaptation of accessible commercial systems. The result is a shared technological trajectory driven by different mechanisms of innovation and support.

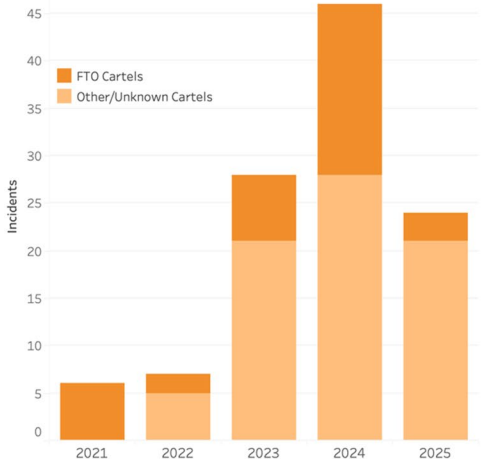
By contrast, unbacked FTOs such as AQAP, the PKK and JNIM have exhibited a far more limited use of unmanned aerial systems (UAS) in both scale and effectiveness. Their operations remain constrained by the lack of state sponsorship and limited access to military-grade platforms. However, these groups, especially ISIS- and al-Qaeda–-affiliated groups, demonstrated a more gradual but consistent increase in drone use over time (see Figure 2). Rather than short-lived surges, these groups exhibit incremental integration of drones into insurgent operations, most notably in Africa, especially from 2023 onward.

Cartel adoption of drones follows a different trajectory. Their drone use emerged relatively later and remained episodic, characterized by short-duration, localized campaigns rather than consistent operational tempos. Although the number of cartel actors employing drones have increased over time, particularly after 2022, overall activity remained lower in volume than that of terrorist organizations. FTO-designated cartels,



Iran-backed..	Actor1 (group) 2	Incidents	Fatalities
Yes	Houthis	633	412
	Hezbollah	371	14
	Islamic Resistance in Iraq	103	13
	Islamic Revolutionary Guard Corps	23	31
	Hamas Movement	14	6
	Total	**1,144**	**476**
No	PKK: Kurdistan Workers Party	74	1
	JNIM: Group for Support of Islam an..	68	30
	FARC-Dissidents	54	3
	AQAP: Al Qaeda in the Arabian Penin..	34	2
	TTP: Tehreek-i-Taliban Pakistan	31	5
	FARC Dissidents	19	4
	ELN: National Liberation Army	13	18
	HTS: Hayat Tahrir al Sham	8	2
	Islamic State West Africa Province (I..	5	22
	Islamic State Somalia Province	5	2
	Islamic State West Africa Province (I..	3	2
	LeT: Lashkar-e-Taiba	1	0
	Islamic State in Iraq and the Levant (..	1	0
	Total	**316**	**91**
Grand Total		**1,460**	**567**

Figure 2. Trends of drone attacks by terrorist groups, 2020–2025.[114]



FTO-Designated..	Actor1	Incidents	Fatalities
Yes	La Familia Michoacana Cartel	25	1
	CJNG: Jalisco New Generation Cartel	11	0
	Total	36	1
No	Los Ardillos Gang	5	0
	Unidentified Gang (Ecuador)	4	0
	Los Viagras Gang	4	0
	CV: Red Command	2	0
	Los Escorpiones Gang	1	3
	CMNG: Michoacan New Generation Cartel	1	0
	Unidentified Armed Group (Mexico)	43	9
	Unidentified Gang (Mexico)	15	2
	Total	75	14
Grand Total		111	15

Figure 3. Trends of drone attacks by cartels, 2020–2025.[115]

specifically La Nueva Familia Michoacana (LNFM; 25 incidents) and the Cártel de Jalisco Nueva Generación (CJNG; 11 incidents), account for the majority of recorded cartel drone incidents. It is also important to note that ACLED records 56 drone-related incidents attributed to unidentified armed groups, some of which may plausibly involve these FTO-designated cartels (see Figure 3).

In terms of **geographic dispersion**, terrorist drone operations exhibit broad spatial reach across multiple conflict zones. Iran-backed groups operate across Yemen, Iraq, Syria, Lebanon, Israel, the Red Sea and the Gulf of Aden, which reflect access to long-range platforms and external logistical support. Unbacked terrorist groups display more geographically clustered patterns, with drone use concentrated in active insurgent theaters such as the Sahel, the Lake Chad Basin, Somalia and parts of Pakistan and Colombia (see Figure 2). Unlike terrorist organizations, cartel drone activity is highly localized, overwhelmingly concentrated in western and southwestern Mexico, with incidents clustering around contested municipalities and rural corridors.

These geographic patterns underscore a crucial distinction between technological convergence and strategic convergence. Although terrorists and cartels increasingly employ similar drone technologies, they do so within different operational environments. Terrorist organizations project force across regional conflict theaters, whereas cartel drone operations remain closely tied to localized contests over territory, trafficking routes and criminal governance.

Civilian targeting further differentiates actor behavior (see Figure 4). Cartels, both FTO-designated (74%) and non-designated (53%), employ drones against civilians at substantially higher rates than terrorist organizations. Over half of all cartel drone incidents involve civilian targets, including community defense groups, villages, homes and civilian infrastructure, aligning with cartel strategies of forced displacement and public intimidation. In contrast, terrorist organizations overwhelmingly direct drone attacks against non-civilian targets, particularly military forces and security installations. Iran-backed and unbacked terrorist groups exhibit significantly lower rates of civilian

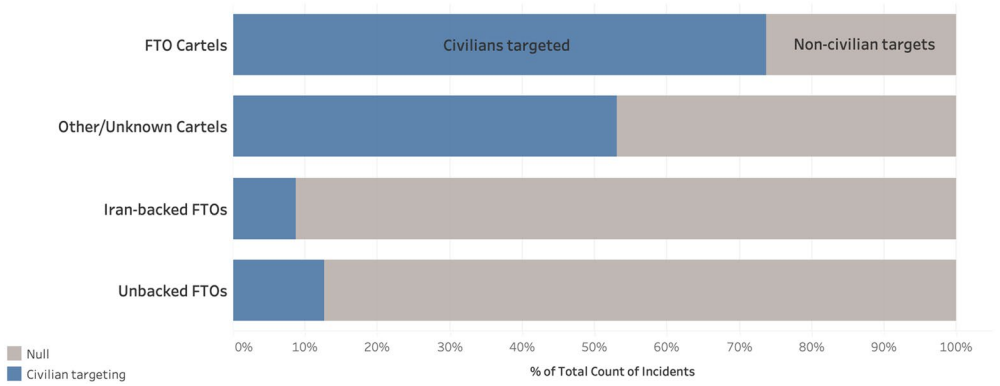


Figure 4. Civilian targeted incidents by groups.[116]

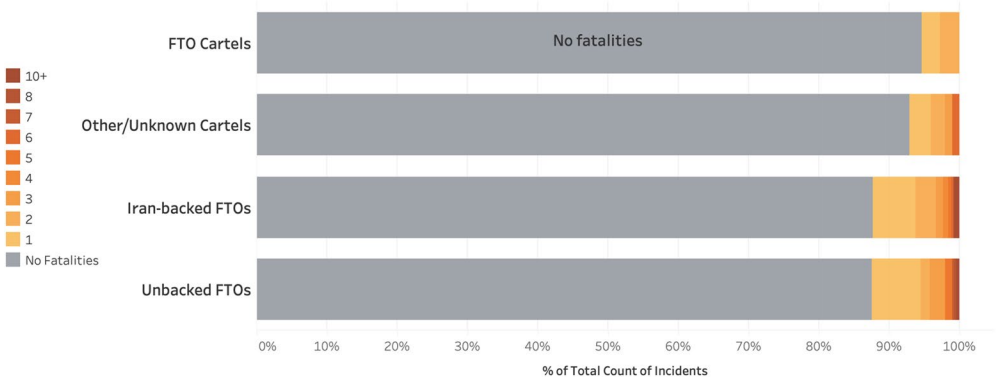


Figure 5. Lethality of incidents by groups.[117]

targeting (9% and 12%, respectively). Their drone-involved attacks primarily targeted state security forces and infrastructures.

In terms of **lethality**, the vast majority of drone-involved incidents do not result in fatalities (see Figure 5). Across all actors, 89% of drone-related incidents recorded in ACLED report zero fatalities, compared to 47% of non-drone attacks by the same pool of actors. However, drone attacks conducted by terrorist organizations, both Iran-backed (12% fatal) and unbacked (11% fatal), are significantly more likely to result in fatalities than those carried out by FTO-designated cartels (5%) and other cartels (7%). This difference aligns with terrorist groups' greater access to more sophisticated drones with larger payloads and longer ranges, as well as their more frequent use of drones as part of combined-arms operations.

For example, JNIM-related incidents recorded higher casualty counts than those of other groups, this primarily reflects its use of drones in combined-arms raids, for intelligence, surveillance, reconnaissance (ISR) and payload delivery, rather than as standalone stand-off strikes. The elevated fatalities in these attacks largely resulted from coordinated ground tactics, including raids and firearm assaults, with drones serving an enabling role and force multiplier role.

Taken together, these findings demonstrate that technological convergence among violent non-state actors is occurring without complete organizational convergence. Cartels increasingly employ drones in ways once associated primarily with insurgent and terrorist organizations, including precision strikes, coordinated attacks and sustained aerial intimidation. At the same time, terrorist organizations continue to exhibit distinct operational patterns shaped by ideological objectives, external sponsorship and participation in armed conflict. These findings suggest that traditional distinctions between terrorism and organized crime remain valuable for understanding strategic intent but are becoming less useful for understanding technological adoption and operational capability.

Discussion

This study demonstrates the expanding and increasingly consequential use of UAS by both terrorist organizations and cartels, while underscoring important differences in how these actors adopt and use drones.

Rather than suggesting that terrorism and organized crime are becoming indistinguishable, our findings indicate that these actors are converging technologically while remaining strategically distinct. This distinction is important because it suggests that future counter-drone policies should focus not only on actor identities but also on the shared technological ecosystems through which violent non-state actors acquire, adapt and operationalize drone capabilities.

Although both increasingly rely on drones for ISR and kinetic attacks, they do so with markedly different levels of strategic intent, technical sophistication and external support. While the VNSA Technology Adoption Curve provides a useful framework for conceptualizing these dynamics, our combined literature review and event-based analysis reveal substantial variation in adaptation trajectories across groups. We summarize the key distinctions and overlap between FTOs and cartel/TCOs across the adoption and the use of drones in Appendix B, Table B1.

At the same time, these findings should be interpreted in light of the limitations inherent in open-source data collections. Event datasets such as ACLED and GRID do not capture the full universe of drone-involved activity by terrorist organizations and cartels and are subject to both random and systematic biases. Reporting intensity varies across regions and actors and media-driven coverage may overrepresent highly newsworthy incidents while undercounting failed attacks or events occurring in areas with limited media access.[118] In addition, a subset of drone attacks is attributed to unknown perpetrators, particularly in cases where groups do not claim responsibility or where incidents lack distinctive terrorist group or cartel patterns in terms of tactics or location. As a result, the observed trends likely represent conservative estimates of drone use and should be interpreted as indicative of broader patterns rather than precise measures of operational scale.

Overall, convergence is most evident among cartels and non–state-sponsored terrorist organizations, which largely cluster in the iteration to early breakthrough stages. These actors rely heavily on commercially available drones, improvised modifications and rapid tactical learning to support localized operations. Divergence is driven primarily by differences in external support and strategic objectives. Iran-backed organizations, benefited from state sponsorship, have progressed further into the late breakthrough and early competition stages. By contrast, unbacked terrorist groups and cartels remain

largely constrained to tactical-level adoption, employing drones primarily as force multipliers for disruption, battlefield coordination and paramilitary-style operations rather than as instruments of sustained aerial warfare.

Our trend analysis further highlights the decisive role of state sponsorship in accelerating technological adoption. This acceleration reflects the role of access to military-grade platforms with extended range and payload capacity, as well as stable logistical, training and technical support networks. At the same time, ISIS- and al-Qaeda–affiliated organizations, particularly in Africa, have demonstrated persistent, bottom-up adaptation, facilitated by transnational knowledge diffusion, intra-network expertise sharing and weak state security forces. These groups have sustained consistent growth in drone use over time.

Cartels follow a distinct but increasingly concerning trajectory. Their adoption of drones began later and remains more episodic, characterized by short-duration, geographically localized campaigns. However, groups such as La Nueva Familia Michoacana (LNFM) and the Cártel de Jalisco Nueva Generación (CJNG) have advanced rapidly into the early breakthrough stage. These cartels have integrated drones for ISR, explosive delivery and, in some cases, chemical dispersal. This pattern suggests that cartel drone adoption is becoming more institutionalized and operationally significant.

Patterns of civilian targeting and lethality further differentiate these actors. Cartels employ drones against civilian targets at substantially higher rates than terrorist organizations, which reflects strategies of forced displacement and public intimidation. Terrorist organizations, particularly Iran-backed groups, overwhelmingly direct drone attacks against security targets and critical infrastructures. Although terrorist drone attacks are more likely to result in fatalities than cartel attacks, overall lethality remains relatively low, especially after October 7. One explanation for this pattern is that many Iran-backed drone and missile attacks were directed against hardened U.S. and Israeli military targets and were intercepted by advanced air and missile defense systems. While these defenses significantly reduced casualties, they imposed substantial economic costs. These incidences reveal persistent economic asymmetry between the relatively low cost of offensive drones and the high expense of counter-UAS and air-defense measures.

From a policy perspective, these dynamics underscore the urgency of developing integrated counter-drone strategies that transcend traditional threat classifications. Existing frameworks often separate counterterrorism and counternarcotics efforts, overlooking the converging technological behaviors of terrorist and criminal actors. Effective responses must therefore address both shared enablers, such as commercial drone supply chains and knowledge diffusion and actor-specific pathways of adaptation. For state-sponsored terrorist organizations, policy priorities should emphasize counter-proliferation, sponsorship interdiction and disruption of military-grade UAS supply networks. For cartels and other criminal actors, the emphasis should shift toward supply-chain regulation, market controls on high-capacity commercial drones and embedded technical safeguards (eg firmware controls and geofencing). Strengthening trade controls and supply-chain oversight for high-capacity FPV components, remote video transmitters, payload-release mechanisms and dual-use avionics could increase acquisition costs for both terrorist organizations and cartels.[119,120]

More broadly, drones continue to reshape asymmetric warfare by extending the reach of violent non-state actors and eroding the state's monopoly over the air domain.

As terrorist groups and cartels increasingly converge in drone capabilities and tactics, the boundary between ideological and criminal violence becomes progressively blurred. Addressing this challenge will require integrated, cost-effective counter-drone strategies that harden vulnerable targets, disrupt drone ecosystems and prevent the proliferation of aerial violence by both terrorist and criminal organizations.

Notes

1. The quotation is drawn from a practitioner focus group convened by the authors on 20 February 2025, as part of an NCITE-sponsored research project examining terrorist use of unmanned aerial systems. Participants represented multiple U.S. federal agencies. Individual identities are withheld in accordance with the study's research protocol and Institutional Review Board (IRB) commitments to confidentiality.
2. Here "grey literature" refers to non-peer-reviewed but publicly available sources, including government reports, policy documents, think tank publications that have substantially informed research on terrorism, organized crime, and unmanned aerial systems.
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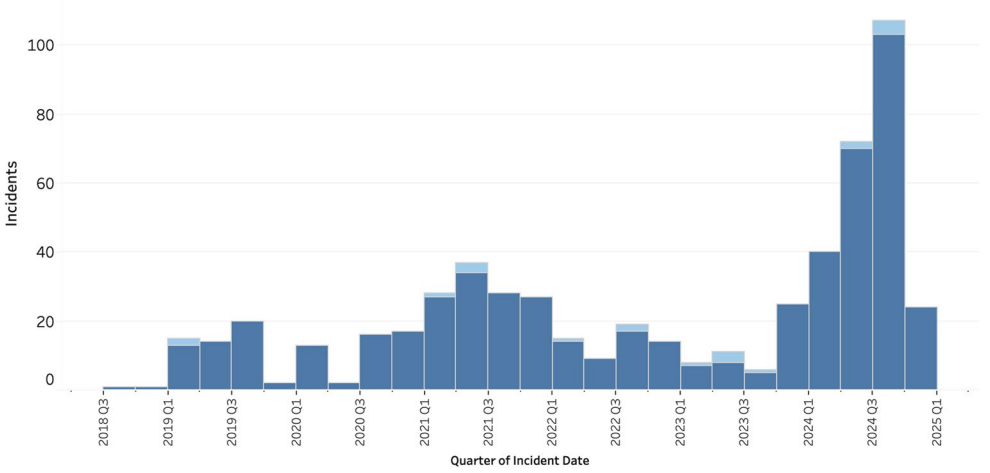
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Appendix A

As a robustness check, we used the Global Terrorism Trends and Analysis Center (GTTAC) Record of Incident Database (GRID), an open-source dataset that has systematically documented global terrorist incidents since 2018.[121] The database compiles open-source intelligence on terrorist activity based on the State Department's definition of terrorism,[122] recording information on incidents, actors, tactics, weapons and targets using verified media sources and data aggregators. The data were retrieved on 15 January 2025 (see Figure 6).

Our findings from the GRID closely align with the findings from ACLED. In both datasets, we observed clear distinctions in the scale and sophistication of drone operations among Iran-backed groups (eg the Houthis, Hezbollah and the Islamic Resistance) compared to non–state-sponsored Foreign Terrorist Organizations (FTOs) such as the Kurdistan Workers' Party



Perpetrator	Incidents	Killed
Houthi extremists	373	260
Hezbollah	164	7
HAMAS	12	2
Islamic Revolutionary Guard Corps	8	20
Kurdistan Workers Party	7	0
AQAP	5	0
Jama'at Nusrat al-Islam wal-Muslimin (JNIM)	3	135
Palestine Islamic Jihad	1	0
National Liberation Army (ELN)	1	5
Grand Total	**573**	**429**

Country	Incidents	Killed	Wounded
YEMEN	193	246	399
SAUDI ARABIA	171	10	104
ISRAEL	170	5	32
LEBANON	11	0	0
IRAQ	9	15	48
TURKEY	5	0	0
EGYPT	4	0	6
MALI	3	135	45
OMAN	2	0	0
GAZA STRIP	2	0	0
VENEZUELA	1	5	0
UNITED ARAB EMIRATES	1	3	6
SYRIA	1	10	0
Grand Total	573	429	640

Figure 6. Drone attacks by FTO-designated terrorist groups, 2018–2024 (source: GRID[123]).

(PKK), al-Qaeda in the Arabian Peninsula (AQAP), Jama'at Nusrat al-Islam wal-Muslimin (JNIM) and the National Liberation Army (ELN). Iran-backed FTOs account for the overwhelming majority of UAS attacks. Of the 573 total UAS incidents recorded between 2018 and 2024, 557 incidents (approximately 97%) were attributed to Iran's proxy groups, including the Houthis (373 incidents), Hezbollah (164) and the Islamic Revolutionary Guard Corps (IRGC) (8). Following 7 October 2023, these groups exhibited a markedly elevated operational tempo. The Houthis alone were responsible for 260 fatalities and 522 injuries resulting from drone-involved attacks. In contrast, while drone use by non–state-sponsored groups (PKK, AQAP, JNIM, PIJ and ELN) remained comparatively limited, it increased incrementally over time. Because the dataset does not include incidents from 2025, we are unable to observe subsequent developments in Africa involving ISIS- and al-Qaeda–affiliated groups during that year. Additionally, GRID does not capture cartel-related violence, precluding direct comparison between terrorist organizations and transnational criminal groups within this dataset.

Appendix B

Table B1. Summary of key insights: terrorist groups vs. cartels.

Dimension	Terrorist organizations	Cartels / TCOs
Strategic motivations	Ideological and political objectives (regime change, policy influence, territorial control).	Profit maximization, territorial and market control, intimidation of rivals, civilians and state forces.
Tactical use of UAS	ISR; stand-off and precision strikes; combined-arms operations; propaganda.	ISR for evasion and territorial monitoring; aerial IED/munition delivery; intimidation and disruption.
Acquisition pathways	State sponsorship (eg Iran); military-grade systems; COTS modification and local assembly.	Primarily COTS drones; commercial modification and improvised weaponization.
Adoption stage (2024–2025)	Iran-backed: late breakthrough → early competition; others: early breakthrough.	Early breakthrough (sustained but improvised integration).
Geographic scope	Regionally expansive (Middle East, Red Sea); insurgent theaters in Africa and Asia.	Highly localized, concentrated in western and southwestern Mexico.
Civilian targeting	Low (≈10%); primarily military and security targets.	High (≈60%); communities, homes and civilian infrastructure.
Lethality	Moderate; higher than cartels but reduced post–Oct. 7 due to air defenses.	Low; few fatalities, effects driven by intimidation and displacement.
Operational role	Integrated into coordinated stand-off and combined-arms campaigns.	Used as localized force multipliers in paramilitary-style operations.